Set
Wed 9 Sep 2026

Problem Set 3 — Functions

Session 3 · Domain, codomain, and undoing

HOW TO ANSWER Every answer carries one sentence of justification. A value on its own is not a solution — say why it is right. You may discuss problems with anyone and may use AI tools, but you must be able to explain every line you submit.

BEFORE YOU START No question here is answered by a verdict alone. "Injective" is not an answer; "injective, because $f(a_1) = f(a_2)$ forces $a_1 = a_2$ since..." is. To *disprove* injectivity or surjectivity you need one concrete witness — to *prove* either, you need an argument covering every element.

Part A · Mechanics

1 IMAGES AND PREIMAGES

Let $f : \mathbb{Z} \to \mathbb{Z}$ be given by $f(n) = 3n - 2$. Compute $f(\{-1, 0, 2\})$, $f^{-1}(\{7\})$ and $f^{-1}(\{5\})$. Explain in one sentence why the third answer is what it is.

SOLUTION $f(\{-1, 0, 2\}) = \{-5, -2, 4\}$. For $f^{-1}(\{7\})$: $3n - 2 = 7$ gives $n = 3$, so the preimage is $\{3\}$. For $f^{-1}(\{5\})$: $3n - 2 = 5$ gives $n = \frac{7}{3}$, which is not an integer, so no element of the domain maps to 5 and $f^{-1}(\{5\}) = \varnothing$. A preimage is allowed to be empty — that is exactly what it means for 5 to lie in the codomain but outside the image.

2 IMAGE AGAINST CODOMAIN

Let $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2 - 4x + 7$. State the image of f and prove your answer. Is f surjective? Justify.

SOLUTION Complete the square: $f(x) = (x - 2)^2 + 3$. Since $(x - 2)^2 \geq 0$ for every real x, we get $f(x) \geq 3$, so the image is contained in $[3, \infty)$. Conversely every $y \geq 3$ is attained, at $x = 2 + \sqrt{y - 3}$. Hence the image is exactly $[3, \infty)$. f is **not** surjective: the codomain is $\mathbb{R}$, and 0 (for instance) is never attained because $f(x) \geq 3 > 0$ for all x.

3 UNIT CIRCLE

Derive each value from the unit circle. For each, state the quadrant and the reference angle before you give the number. A recalled value without that working is not a derivation.

a. $\cos \dfrac{2\pi}{3}$

b. $\sin \dfrac{3\pi}{4}$

c. $\cos \dfrac{7\pi}{6}$

d. $\sin \dfrac{5\pi}{3}$

e. $\tan \dfrac{3\pi}{4}$

SOLUTION (a) $\frac{2\pi}{3}$ is in quadrant II, reference angle $\frac{\pi}{3}$; cosine is the x-coordinate, negative there, so $-\frac{1}{2}$. (b) $\frac{3\pi}{4}$, quadrant II, reference $\frac{\pi}{4}$; sine is the y-coordinate, positive there, so $\frac{\sqrt{2}}{2}$. (c) $\frac{7\pi}{6}$, quadrant III, reference $\frac{\pi}{6}$; cosine negative, so $-\frac{\sqrt{3}}{2}$. (d) $\frac{5\pi}{3}$, quadrant IV, reference $\frac{\pi}{3}$; sine negative, so $-\frac{\sqrt{3}}{2}$. (e) $\tan \frac{3\pi}{4} = \dfrac{\sin(3\pi/4)}{\cos(3\pi/4)} = \dfrac{\sqrt{2}/2}{-\sqrt{2}/2} = -1$.

4 CLASSIFICATION

For each function decide whether it is injective, surjective, both or neither. Prove each positive claim; disprove each negative one with an explicit witness.

a. $f : \mathbb{R} \to \mathbb{R},\ f(x) = 5 - 3x$
b. $f : \mathbb{Z} \to \mathbb{Z},\ f(n) = n^2 - n$
c. $f : \mathbb{R} \to \mathbb{R},\ f(x) = x^3 - 3x$
d. $f : \mathbb{N} \to \mathbb{N},\ f(n) = n^2$
e. $f : \mathbb{R} \setminus \{0\} \to \mathbb{R} \setminus \{0\},\ f(x) = \dfrac{1}{x}$
f. $f : \mathbb{R} \to [-1, 1],\ f(x) = \sin x$
g. $f : \mathbb{Z} \times \mathbb{Z} \to \mathbb{Z},\ f(a, b) = 2a + 4b$
h. $S = \{1, 2, 3\},\ f : \mathcal{P}(S) \to \mathcal{P}(S),\ f(X) = S \setminus X$

SOLUTION (a) **Bijective.** Injective: $5 - 3a = 5 - 3b \Rightarrow a = b$. Surjective: given y, take $x = \frac{5-y}{3}$.
(b) **Neither.** Not injective: $f(0) = f(1) = 0$. Not surjective: $n^2 - n = n(n-1)$ is a product of consecutive integers, hence even, so the odd value 1 is never attained.
(c) **Surjective, not injective.** Not injective: $f(0) = 0$ and $f(\sqrt{3}) = 3\sqrt{3} - 3\sqrt{3} = 0$. Surjective: f is continuous with $f(x) \to \pm\infty$ as $x \to \pm\infty$, so by the intermediate value theorem every real is attained.
(d) **Injective, not surjective.** Injective: for $a, b \geq 1$, $a^2 = b^2 \Rightarrow a = b$. Not surjective: 2 is not a perfect square.
(e) **Bijective.** $f(f(x)) = x$ on the whole domain, so f is its own two-sided inverse, and by the theorem a map with a two-sided inverse is a bijection.
(f) **Surjective, not injective.** Not injective: $\sin 0 = \sin \pi = 0$. Surjective: the image of $\sin$ is exactly $[-1, 1]$, which here *is* the codomain — note how shrinking the codomain created surjectivity.
(g) **Neither.** Not injective: $f(2, 0) = 4 = f(0, 1)$. Not surjective: $2a + 4b$ is always even, so 1 is never attained.
(h) **Bijective.** $f(f(X)) = S \setminus (S \setminus X) = X$, so f is its own inverse, hence a bijection.

5 COMPOSITION

Let $f : A \to B$ and $g : B \to C$.

a. Prove that if $g \circ f$ is surjective, then g is surjective.

b. Show by explicit example that f need *not* be surjective under the same hypothesis.

c. In one sentence, say why (a) is about g and not f, given that in class we proved $g \circ f$ injective forces f injective.

> **SOLUTION** (a) Let $c \in C$. Since $g \circ f$ is surjective there is $a \in A$ with $g(f(a)) = c$. Put $b = f(a) \in B$; then $g(b) = c$. As c was arbitrary, g is surjective.
> (b) $A = \{1\}$, $B = \{1, 2\}$, $C = \{1\}$, with $f(1) = 1$ and $g(1) = g(2) = 1$. Then $g \circ f$ is surjective onto C, but f misses 2.
> (c) Surjectivity is a statement about what comes *out*, so it is inherited by the last map in the chain; injectivity is a statement about what goes *in*, so it is inherited by the first.

6 RESTRICTION

Take the rule $x \mapsto x^2 - 4x + 7$ from question 2. Choose a domain $A \subseteq \mathbb{R}$ and a codomain B making $f : A \to B$ a **bijection**, and prove both halves.

> **SOLUTION** Take $A = [2, \infty)$ and $B = [3, \infty)$; write $f(x) = (x-2)^2 + 3$. Injective: for $x_1, x_2 \geq 2$ the quantities $x_1 - 2, x_2 - 2$ are ≥ 0, and $(x_1 - 2)^2 = (x_2 - 2)^2$ with both bases non-negative gives $x_1 - 2 = x_2 - 2$. Surjective: given $y \geq 3$, put $x = 2 + \sqrt{y - 3}$, which lies in A and satisfies $f(x) = y$. ($A = (-\infty, 2]$ with the same B works equally well.)

7 FIND THE ERROR

The following was produced by a chatbot. It contains **three** distinct errors. Find all three; for each, say what is wrong and give a witness where one is needed.

> **Claim.** $f : \mathbb{R} \to \mathbb{R}$, $f(x) = \dfrac{x}{x^2+1}$, is a bijection, and its inverse is $f^{-1}(x) = \dfrac{x^2+1}{x}$.
> **"Proof."** f is injective because it is a ratio of polynomials with no repeated factors. It is surjective because its domain and codomain are both $\mathbb{R}$. Inverting a fraction gives the inverse function.

SOLUTION **Error 1 — not injective.** "Ratio of polynomials with no repeated factors" is not a criterion for anything. Witness: $f(2) = \frac{2}{5}$ and $f\left(\frac{1}{2}\right) = \frac{1/2}{5/4} = \frac{2}{5}$, so $f(2) = f\left(\frac{1}{2}\right)$ with $2 \neq \frac{1}{2}$.

Error 2 — not surjective. Domain and codomain being equal says nothing about surjectivity. In fact $(x \mp 1)^2 \geq 0$ gives $x^2 + 1 \geq \pm 2x$, so $|f(x)| \leq \frac{1}{2}$ for every x; the image is $\left[-\frac{1}{2}, \frac{1}{2}\right]$ and 1 is never attained.

Error 3 — inverse against reciprocal. $\frac{x^2+1}{x}$ is $\frac{1}{f(x)}$, not f^{-1}. Undoing a function is unrelated to flipping a fraction: f^{-1} satisfies $f^{-1}(f(x)) = x$, and here $\frac{1}{f(f(x))} \neq x$ in general. Since f is not a bijection, no inverse function exists at all.

8 REWRITE PRECISELY

Each sentence is sloppy or false as written. Rewrite each so that it is unambiguous and true.

a. "x^2 is not injective."

b. "Every function has an inverse — you just swap x and y."

c. "$\sin^{-1}(x)$ means $\dfrac{1}{\sin x}$."

WHAT A GOOD ANSWER DOES (a) A rule is not a function: e.g. "the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, is not injective, since $f(-1) = f(1)$ but $-1 \neq 1$." Any answer that supplies a domain and codomain and a witness is acceptable.
(b) "A function has an inverse if and only if it is a bijection. Swapping x and y in the equation of the graph yields a function precisely in that case."
(c) "$\sin^{-1}$ denotes the inverse of $\sin$ restricted to $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, where it is a bijection onto $[-1, 1]$. The reciprocal $\frac{1}{\sin x}$ is a different function, written $\csc x$."